Healthcare Analysis Report

1. Project Overview

The Healthcare Emergency Room Analysis Dashboard was created using Power BI to analyze hospital patient records and gain valuable insights into patient admissions, waiting times, demographics, department referrals, and patient satisfaction.

The main objective of this project is to transform raw healthcare data into an interactive dashboard that enables healthcare professionals to monitor emergency room performance, identify trends, and make data-driven decisions.

The complete project was developed using only Power BI.

2. Dataset Summary

Dataset Information

- Total Records: 9,216 patients
- Original Columns: 12 columns

Additional Columns Created in Power BI

- **Full Name –** Combined Patient First Initial and Patient Last Name
- **Date Columns** – Created Year, Month Number, Month Name, and Day columns for time-based analysis

Key Features in Dataset

Patient Details

- Patient ID
- Full Name
- Gender
- Age
- Race

Hospital Performance Details

- Department Referral
- Patient Admission Status
- Patient Waiting Time
- Patient Satisfaction Score

Time-Based Details

- Patient Admission Date
- Monthly Patient Trends
- Year-wise Analysis

Tools Used

- Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Interactive Dashboard Design

3. Data Preparation in Power BI

Data transformation and preparation were performed using Power Query Editor within Power BI.

The following steps were carried out:

- Imported the healthcare Excel dataset into Power BI.
- Checked all column data types and corrected them where required.
- Created a Full Name column by merging Patient First Initial and Patient Last Name.
- Extracted date-related columns from Patient Admission Date to perform monthly and yearly analysis.
- Handled blank values in Department Referral and Patient Satisfaction Score fields where necessary.
- Removed unnecessary formatting issues and prepared the data for dashboard visualization.

4. Calculated Columns Created

Full Name:

```
Full Name =
[Patient First Initial] & " " & [Patient Last Name]
```

Year:

```
Year =
YEAR([Patient Admission Date])
```

5. DAX Measures Used for KPI Cards

Total Patients:

Total Patients =

COUNT('Hospital ER_Data'[Patient ID])

Purpose: Displays the total number of patients who visited the emergency room.

Average Waiting Time:

Average Waiting Time =

AVERAGE('Hospital ER_Data'[Patient Wait Time])

Purpose: Calculates the average amount of time patients wait before receiving medical attention.

Average Patient Satisfaction Score:

Average Patient Satisfaction Score =

AVERAGE('Hospital ER_Data'[Patient Satisfaction Score])

Purpose: Measures the overall patient experience and service quality.

Total Admissions:

Total Admissions =

CALCULATE(

COUNT('Hospital ER_Data'[Patient ID]),

'Hospital ER_Data'[Admission Status] = "Admitted")

Purpose: Displays the total number of patients admitted to the hospital.

6. Dashboard Components in Power BI

KPI Cards

The dashboard contains **five KPI cards** that provide a quick summary of emergency room performance and hospital operations.

The KPI cards display:

- **Total Patients** – Shows the total number of patients who visited the emergency room.
- **Average Waiting Time (Minutes)** – Displays the average waiting time of patients before receiving treatment.
- **Average Satisfaction Score** – Represents the overall patient satisfaction level.
- **Total Departments** – Shows the number of hospital departments involved in patient referrals.

- **Case Management Cases** – Displays the total number of patients requiring case management.

These KPI cards help users quickly understand the overall performance and important healthcare metrics.

Slicers Used

The dashboard includes **four interactive slicers** that allow users to filter the data dynamically.

The slicers are:

- **Patient Gender** – Filters records based on male, female, or other patient categories.
- **Patient Age** – Allows analysis of patients across different age groups.
- **Department Referral** – Filters data based on the department to which patients were referred.
- **Month** – Enables monthly trend analysis by filtering records according to the selected month.

These slicers make the dashboard interactive and allow users to perform customized analysis based on specific criteria.

7. Visualizations Included

The dashboard contains **five analytical charts** and **one detailed data table**.

1. Department-wise Referral Count

A **horizontal bar chart** is used to display the number of patients referred to different hospital departments, helping identify departments with the highest patient volume.

2. Patient Distribution by Gender

A **donut chart** is used to compare the proportion of patients across different gender categories.

3. Monthly Record Count

A **column chart** is used to analyze the number of patient records across different months, helping identify monthly patient admission trends.

4. Department-wise Satisfaction Score

A **bar chart** is used to compare patient satisfaction scores among different departments and evaluate service quality.

5. Patient Details Analysis

A **table visual** is used to display detailed patient information, including full name, department referral, gender, race, age, waiting time, and satisfaction score.

8. Key Insights

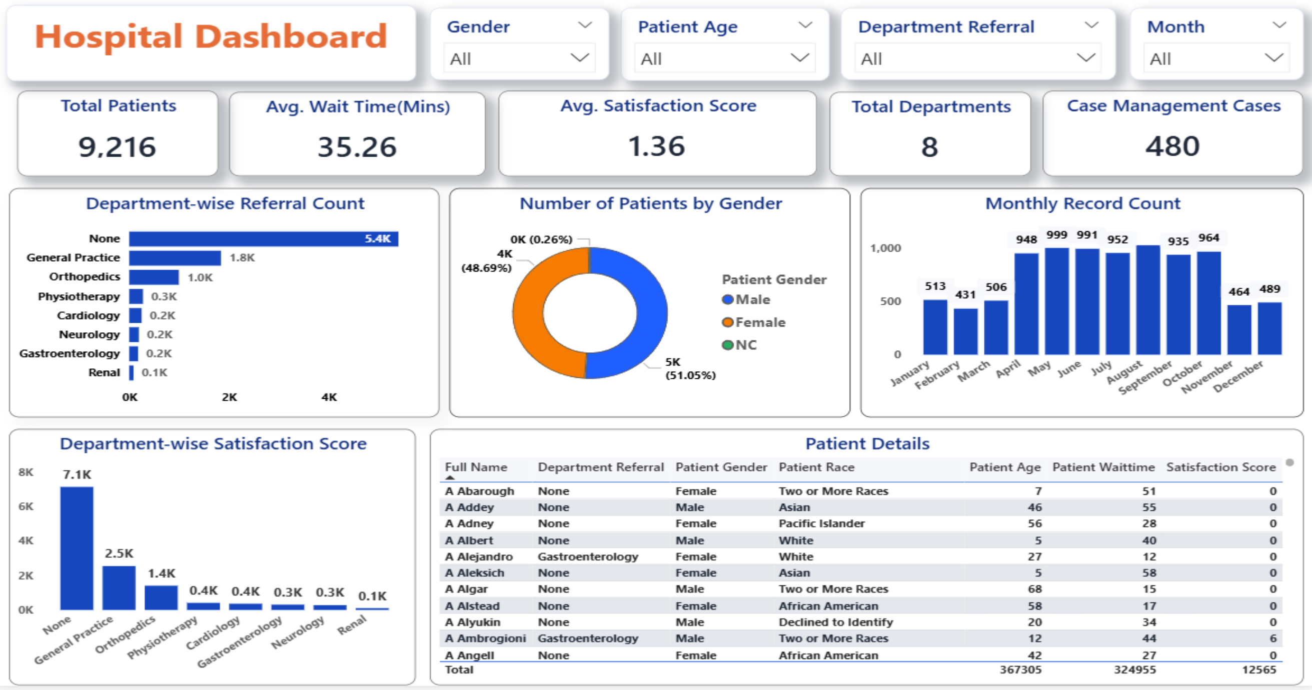
Key Insights

1. **Patient Growth**
   - Total patients increased by 11.9% from March to April.
   - This indicates a rise in patient visits or registrations during April.
2. **Highest Waiting Time**
   - The Neurology department has the highest average waiting time at 55 minutes.
   - This may be due to high patient demand, limited staff availability, or scheduling challenges.
3. **Highest Patient Satisfaction**
   - The Orthopedics department has the highest satisfaction score of 4.6 out of 5.
   - This reflects a positive patient experience and quality service.

Additional Dashboard Highlights

- Total Patients: 9,216
- Average Waiting Time: 35.3 minutes
- Average Satisfaction Score: 4.2 / 5
- Total Departments: 8
- Total Case Management Records: 480

Hospital Performance Dashboard



9. Business Recommendations

- Increase staff availability in departments with higher waiting times, especially Neurology, to reduce patient waiting periods.
- Allocate additional resources and medical personnel to General Practice due to its high patient referral rate.
- Continuously monitor patient satisfaction metrics to maintain and improve healthcare service quality.
- Use monthly patient trend analysis for effective staff scheduling, resource planning, and emergency room management.
- Analyze demographic patterns such as age, gender, and race to provide more personalized and efficient healthcare services.

10. Project Conclusion

This Healthcare Emergency Room Analysis project demonstrates the ability to perform data transformation, create calculated columns and DAX measures, design interactive dashboards, and generate meaningful business insights using Power BI. The dashboard provides a clear overview of hospital performance and helps support better operational and strategic decision-making.